

Digital Signal Processing

Final Exam

(EE3031)

Date: May 27th 2025

Course Instructor(s)

Dr. Muhammad Akmal

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 5

Roll No

Section

Student Signature

Attach the question paper with answer sheet.
Do not write below this line.

Attempt all the questions.

CLO # 5: Design filters using various techniques

Q1:

[10]

A digital low-pass filter is required to meet the following specifications:

Passband ripple: ≤ 2 dB

Passband edge: 6 kHz

Stopband attenuation: ≥ 50 dB

Stopband edge: 9 kHz

Sample rate: 34 kHz

The filter is to be designed by performing a bilinear transformation on an analog system function. Determine what order i) Butterworth, and ii) Chebyshev I analog designs must be used to meet the specifications in the digital implementation.

CLO # 3: Apply z-transform to analyze discrete time systems

Q2: Determine the z-transform of the following signals:

[10]

a) $x(n) = a^n (\cos w_0 n) u(n)$

b) $x(n) = a^n (\sin w_0 n) u(n)$

CLO # 5: Design filters using various techniques

Q3:

[10]

Design an FIR Bandstop digital filter approximating the following frequency response:

$$H_d(\omega) = \begin{cases} 1 & \text{for } |\omega| \leq \frac{\pi}{6} \\ 0 & \text{for } \frac{\pi}{6} < |\omega| \leq \frac{\pi}{3} \\ 1 & \text{for } \frac{\pi}{3} \leq |\omega| \leq \pi \end{cases}$$

- a) Determine the coefficients of a 5-tap filter based on the window method with a i) Hanning window, and ii) Blackman window.
(b) Determine and plot the magnitude, and phase response of the filter.

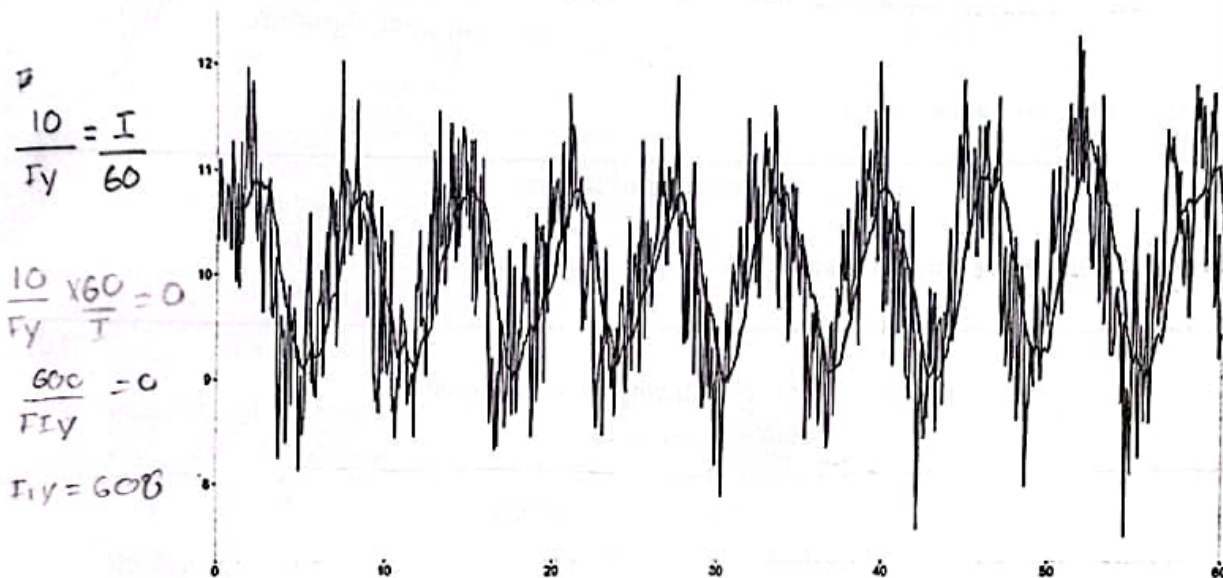
CLO # 3: Apply z-transform to analyze discrete time systems

Q4:

[10]

An audio processing system samples an analog signal at 10 kHz, and uses a digital filter to remove 60 Hz power-line noise. The input-output relation of the system is as follows:

$$y[n] = 0.8y[n-1] + x[n] - 0.9x[n-1]$$



where:

$x[n]$ = input signal

$y[n]$ = filtered output

Determine the transfer function $H(z)$ of the filter using the Z-transform. Also show magnitude, and Pole-Zero locations.

CLO # 5: Design filters using various techniques

Q5:

[10]

Design a type I Low Pass Chebyshev filter that has a 1-dB ripple in the passband, a cutoff frequency $\Omega_p = 1000\pi$, a stopband frequency of 2000π , and an attenuation of 40dB or more for $\Omega \geq \Omega_s$.

Instructions:

1. Make a word file with all codes and plots in it.
2. Do not forget to write your Name and Roll Numbers on it.
3. No need to write anything on this paper.

Question No.1 (LLO No. 4)

Marks: 10

i) Build MATLAB code for the following filters:

- High Pass Butterworth filter
- High Pass Chebyshev I filter

Where,

1. Order of filter = $N = 7$
2. p-p ripple = $R = 0.5$ dB,
3. Cut-off frequency = choose by yourself.

Display magnitude and phase plots.

Question No.2 (LLO No. 4)

Marks: 10

Implement the designed filters on the following signal:

`input_signal = randn(1000,1);`

Display plots of input signal and filtered signal.